20. Two Wire Interface (TWI)

This section describes the 2-wire interface. The 2-wire bus is a bi-directional 2-wire serial communication standard. It is designed primarily for simple but efficient integrated circuit (IC) control. The system is comprised of two lines, SCL (Serial Clock) and SDA (Serial Data) that carry information between the ICs connected to them. The serial data transfer is limited to 400 Kbit/s in standard mode. Various communication configuration can be designed using this bus. Figure 20-1 shows a typical 2-wire bus configuration. All the devices connected to the bus can be master and slave.

Figure 20-1. 2-wire Bus Configuration

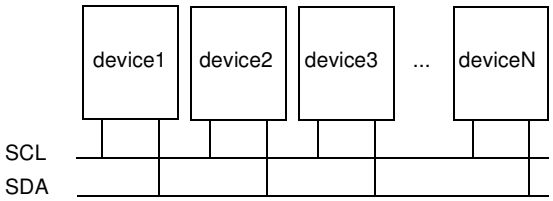
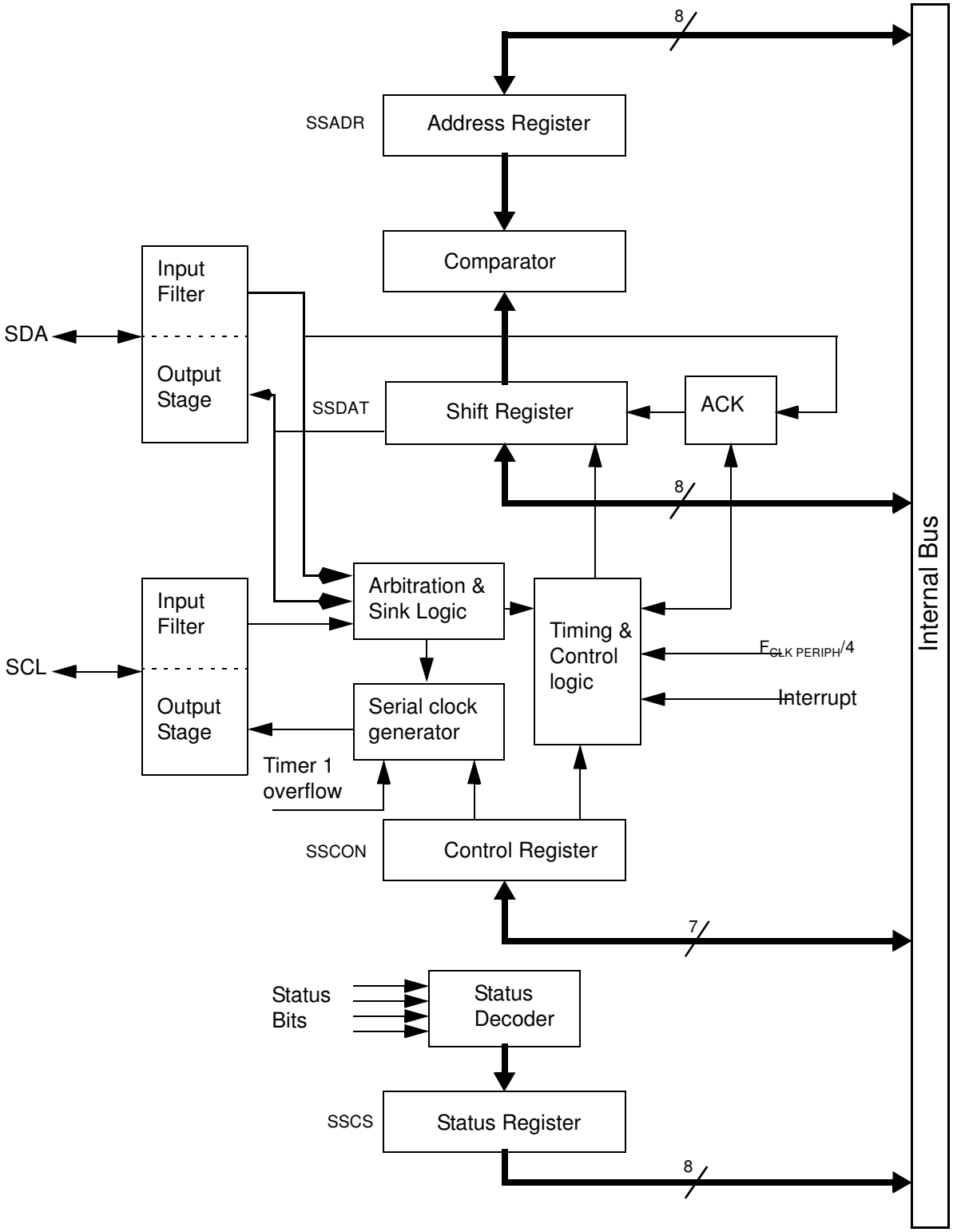


Figure 20-2. Block Diagram



20.1 Description

The CPU interfaces to the 2-wire logic via the following four 8-bit special function registers: the Synchronous Serial Control register (SSCON; Table 20-10), the Synchronous Serial Data register (SSDAT; Table 20-11), the Synchronous Serial Control and Status register (SSCS; Table 20-12) and the Synchronous Serial Address register (SSADR Table 20-13).

SSCON is used to enable the TWI interface, to program the bit rate (see Table 20-3), to enable slave modes, to acknowledge or not a received data, to send a START or a STOP condition on the 2-wire bus, and to acknowledge a serial interrupt. A hardware reset disables the TWI module.

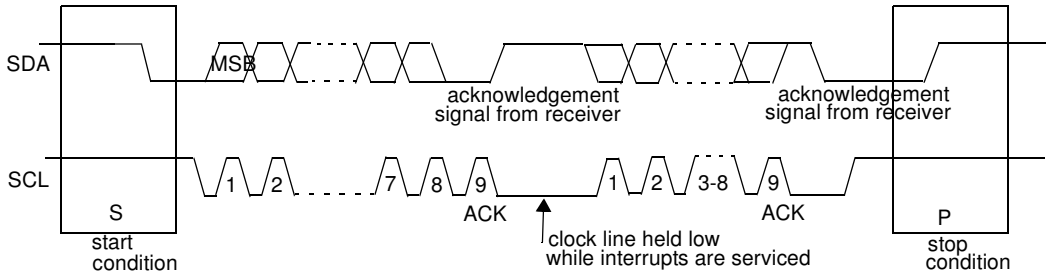
SSCS contains a status code which reflects the status of the 2-wire logic and the 2-wire bus. The three least significant bits are always zero. The five most significant bits contains the status code. There are 26 possible status codes. When SSCS contains F8h, no relevant state information is available and no serial interrupt is requested. A valid status code is available in SSCS one machine cycle after SI is set by hardware and is still present one machine cycle after SI has been reset by software. to Table 20-9. give the status for the master modes and miscellaneous states.

SSDAT contains a byte of serial data to be transmitted or a byte which has just been received. It is addressable while it is not in process of shifting a byte. This occurs when 2-wire logic is in a defined state and the serial interrupt flag is set. Data in SSDAT remains stable as long as SI is set. While data is being shifted out, data on the bus is simultaneously shifted in; SSDAT always contains the last byte present on the bus.

SSADR may be loaded with the 7-bit slave address (7 most significant bits) to which the TWI module will respond when programmed as a slave transmitter or receiver. The LSB is used to enable general call address (00h) recognition.

Figure 20-3 shows how a data transfer is accomplished on the 2-wire bus.

Figure 20-3. Complete Data Transfer on 2-wire Bus



The four operating modes are:

- Master Transmitter
- Master Receiver
- Slave transmitter
- Slave receiver

Data transfer in each mode of operation is shown in Table to Table 20-9 and Figure 20-4. to Figure 20-7.. These figures contain the following abbreviations:

S : START condition

R : Read bit (high level at SDA)

W : Write bit (low level at SDA)

A: Acknowledge bit (low level at SDA)

$\overline{A}$: Not acknowledge bit (high level at SDA)

Data: 8-bit data byte

P : STOP condition

In Figure 20-4 to Figure 20-7, circles are used to indicate when the serial interrupt flag is set. The numbers in the circles show the status code held in SSCS. At these points, a service routine must be executed to continue or complete the serial transfer. These service routines are not critical since the serial transfer is suspended until the serial interrupt flag is cleared by software.

When the serial interrupt routine is entered, the status code in SSCS is used to branch to the appropriate service routine. For each status code, the required software action and details of the following serial transfer are given in Table to Table 20-9.

20.1.1 Master Transmitter Mode

In the master transmitter mode, a number of data bytes are transmitted to a slave receiver (Figure 20-4). Before the master transmitter mode can be entered, SSCON must be initialised as follows:

Table 20-1. SSCON Initialization

CR2	SSIE	STA	STO	SI	AA	CR1	CR0
bit rate	1	0	0	0	X	bit rate	bit rate

CR0, CR1 and CR2 define the internal serial bit rate if external bit rate generator is not used. SSIE must be set to enable TWI. STA, STO and SI must be cleared.

The master transmitter mode may now be entered by setting the STA bit. The 2-wire logic will now test the 2-wire bus and generate a START condition as soon as the bus becomes free. When a START condition is transmitted, the serial interrupt flag (SI bit in SSCON) is set, and the status code in SSCS will be 08h. This status must be used to vector to an interrupt routine that loads SSDAT with the slave address and the data direction bit (SLA+W).

When the slave address and the direction bit have been transmitted and an acknowledgement bit has been received, SI is set again and a number of status code in SSCS are possible. There are 18h, 20h or 38h for the master mode and also 68h, 78h or B0h if the slave mode was enabled (AA=logic 1). The appropriate action to be taken for each of these status code is detailed in Table . This scheme is repeated until a STOP condition is transmitted.

SSIE, CR2, CR1 and CR0 are not affected by the serial transfer and are referred to Table 7 to Table 11. After a repeated START condition (state 10h) the TWI module may switch to the master receiver mode by loading SSDAT with SLA+R.

20.1.2 Master Receiver Mode

In the master receiver mode, a number of data bytes are received from a slave transmitter (Figure 20-5). The transfer is initialized as in the master transmitter mode. When the START condition has been transmitted, the interrupt routine must load SSDAT with the 7-bit slave

address and the data direction bit (SLA+R). The serial interrupt flag SI must then be cleared before the serial transfer can continue.

When the slave address and the direction bit have been transmitted and an acknowledgement bit has been received, the serial interrupt flag is set again and a number of status code in SSCS are possible. There are 40h, 48h or 38h for the master mode and also 68h, 78h or B0h if the slave mode was enabled (AA=logic 1). The appropriate action to be taken for each of these status code is detailed in Table . This scheme is repeated until a STOP condition is transmitted.

SSIE, CR2, CR1 and CR0 are not affected by the serial transfer and are referred to Table 7 to Table 11. After a repeated START condition (state 10h) the TWI module may switch to the master transmitter mode by loading SSDAT with SLA+W.

20.1.3 Slave Receiver Mode

In the slave receiver mode, a number of data bytes are received from a master transmitter (Figure 20-6). To initiate the slave receiver mode, SSADR and SSCON must be loaded as follows:

Table 20-2. SSADR: Slave Receiver Mode Initialization

A6	A5	A4	A3	A2	A1	A0	GC
own slave address							

The upper 7 bits are the address to which the TWI module will respond when addressed by a master. If the LSB (GC) is set the TWI module will respond to the general call address (00h); otherwise it ignores the general call address.

Table 20-3. SSCON: Slave Receiver Mode Initialization

CR2	SSIE	STA	STO	SI	AA	CR1	CR0
bit rate	1	0	0	0	1	bit rate	bit rate

CR0, CR1 and CR2 have no effect in the slave mode. SSIE must be set to enable the TWI. The AA bit must be set to enable the own slave address or the general call address acknowledgement. STA, STO and SI must be cleared.

When SSADR and SSCON have been initialised, the TWI module waits until it is addressed by its own slave address followed by the data direction bit which must be at logic 0 (W) for the TWI to operate in the slave receiver mode. After its own slave address and the W bit have been received, the serial interrupt flag is set and a valid status code can be read from SSCS. This status code is used to vector to an interrupt service routine.The appropriate action to be taken for each of these status code is detailed in Table . The slave receiver mode may also be entered if arbitration is lost while TWI is in the master mode (states 68h and 78h).

If the AA bit is reset during a transfer, TWI module will return a not acknowledge (logic 1) to SDA after the next received data byte. While AA is reset, the TWI module does not respond to its own slave address. However, the 2-wire bus is still monitored and address recognition may be resume at any time by setting AA. This means that the AA bit may be used to temporarily isolate the module from the 2-wire bus.

20.1.4 Slave Transmitter Mode

In the slave transmitter mode, a number of data bytes are transmitted to a master receiver (Figure 20-7). Data transfer is initialized as in the slave receiver mode. When SSADR and SSCON have been initialized, the TWI module waits until it is addressed by its own slave address followed by the data direction bit which must be at logic 1 (R) for TWI to operate in the slave transmitter mode. After its own slave address and the R bit have been received, the serial interrupt flag is set and a valid status code can be read from SSCS. This status code is used to vector to an interrupt service routine. The appropriate action to be taken for each of these status code is detailed in Table . The slave transmitter mode may also be entered if arbitration is lost while the TWI module is in the master mode.

If the AA bit is reset during a transfer, the TWI module will transmit the last byte of the transfer and enter state C0h or C8h. the TWI module is switched to the not addressed slave mode and will ignore the master receiver if it continues the transfer. Thus the master receiver receives all 1's as serial data. While AA is reset, the TWI module does not respond to its own slave address. However, the 2-wire bus is still monitored and address recognition may be resume at any time by setting AA. This means that the AA bit may be used to temporarily isolate the TWI module from the 2-wire bus.

20.1.5 Miscellaneous States

There are two SSCS codes that do not correspond to a define TWI hardware state (Table 20-9). These codes are discuss hereafter.

Status F8h indicates that no relevant information is available because the serial interrupt flag is not set yet. This occurs between other states and when the TWI module is not involved in a serial transfer.

Status 00h indicates that a bus error has occurred during a TWI serial transfer. A bus error is caused when a START or a STOP condition occurs at an illegal position in the format frame. Examples of such illegal positions happen during the serial transfer of an address byte, a data byte, or an acknowledge bit. When a bus error occurs, SI is set. To recover from a bus error, the STO flag must be set and SI must be cleared. This causes the TWI module to enter the not addressed slave mode and to clear the STO flag (no other bits in SSCON are affected). The SDA and SCL lines are released and no STOP condition is transmitted.

20.2 Notes

The TWI module interfaces to the external 2-wire bus via two port pins: SCL (serial clock line) and SDA (serial data line). To avoid low level asserting on these lines when the TWI module is enabled, the output latches of SDA and SLC must be set to logic 1.

Table 20-4. Bit Frequency Configuration

			Bit Frequency (kHz)		
CR2	**CR1**	**CR0**	**F_{OSCA}= 12 MHz**	**F_{OSCA} = 16 MHz**	**F_{OSCA} divided by**
0	0	0	47	62.5	256
0	0	1	53.5	71.5	224
0	1	0	62.5	83	192
0	1	1	75	100	160

			Bit Frequency (kHz)		
CR2	**CR1**	**CR0**	**F_{OSCA}= 12 MHz**	**F_{OSCA} = 16 MHz**	**F_{OSCA} divided by**
1	0	0	-	-	Unused
1	0	1	100	133.3	120
1	1	0	200	266.6	60
1	1	1	0.5 <. < 62.5	0.67 <. < 83	Timer 1 in mode 2 can be used as TWI baudrate generator with the following formula: 96.(256-”Timer1 reload value”)

Figure 20-4. Format and State in the Master Transmitter Mode

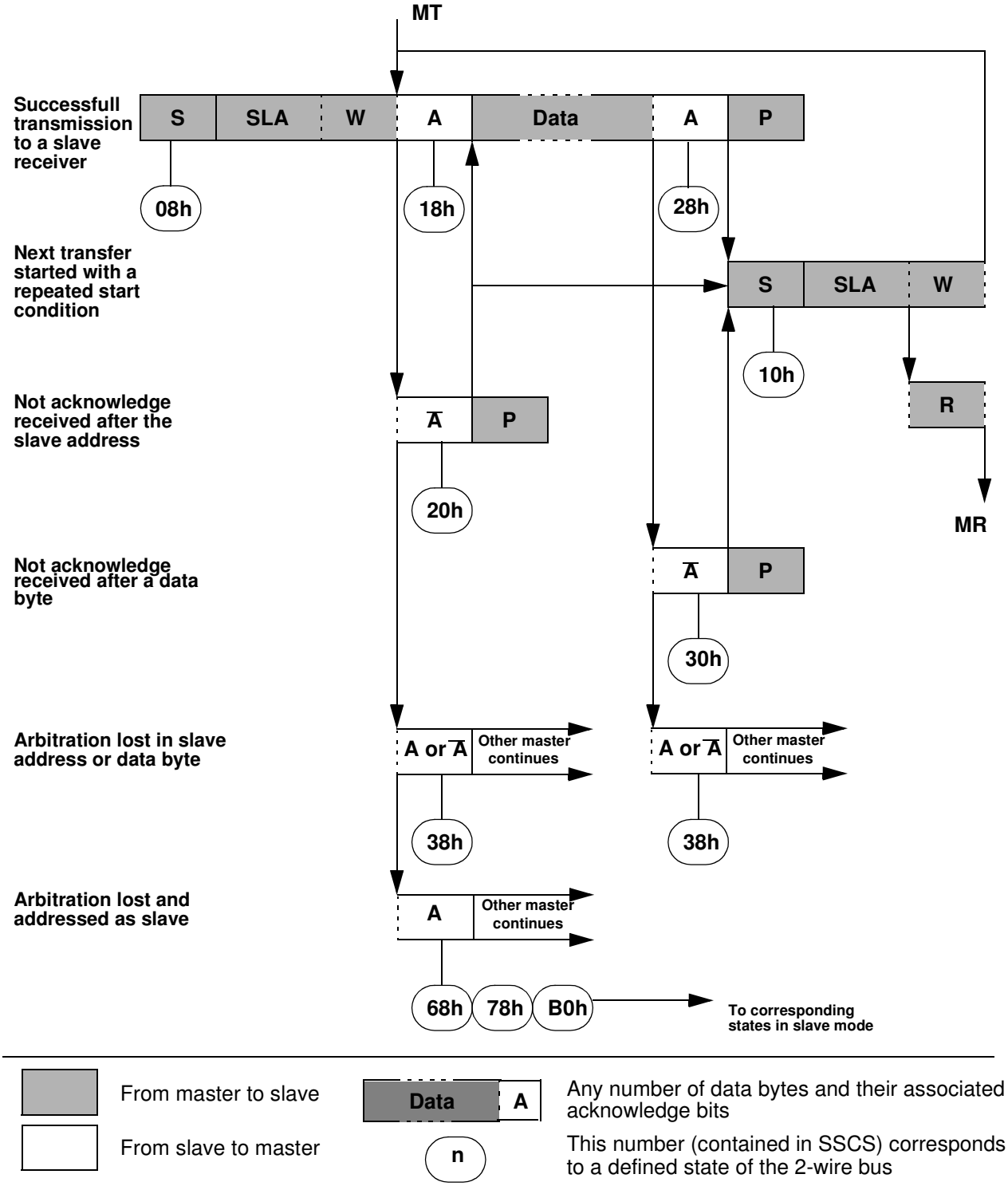


Table 20-5. Status in Master Transmitter Mode

Status Code SSSTA	Status of the Two-wire Bus and Two-wire Hardware	Application software response					Next Action Taken by Two-wire Hardware
		To/From SSDAT	To SSCON				
			SSSTA	SSSTO	SSI	SSAA	
08h	A START condition has been transmitted	Write SLA+W	X	0	0	X	SLA+W will be transmitted.
10h	A repeated START condition has been transmitted	Write SLA+W	X	0	0	X	SLA+W will be transmitted.
		Write SLA+R	X	0	0	X	SLA+R will be transmitted. Logic will switch to master receiver mode
18h	SLA+W has been transmitted; ACK has been received	Write data byte	0	0	0	X	Data byte will be transmitted.
		No SSDAT action	1	0	0	X	Repeated START will be transmitted.
		No SSDAT action	0	1	0	X	STOP condition will be transmitted and SSSTO flag will be reset.
		No SSDAT action	1	1	0	X	STOP condition followed by a START condition will be transmitted and SSSTO flag will be reset.
20h	SLA+W has been transmitted; NOT ACK has been received	Write data byte	0	0	0	X	Data byte will be transmitted.
		No SSDAT action	1	0	0	X	Repeated START will be transmitted.
		No SSDAT action	0	1	0	X	STOP condition will be transmitted and SSSTO flag will be reset.
		No SSDAT action	1	1	0	X	STOP condition followed by a START condition will be transmitted and SSSTO flag will be reset.
28h	Data byte has been transmitted; ACK has been received	Write data byte	0	0	0	X	Data byte will be transmitted.
		No SSDAT action	1	0	0	X	Repeated START will be transmitted.
		No SSDAT action	0	1	0	X	STOP condition will be transmitted and SSSTO flag will be reset.
		No SSDAT action	1	1	0	X	STOP condition followed by a START condition will be transmitted and SSSTO flag will be reset.
30h	Data byte has been transmitted; NOT ACK has been received	Write data byte	0	0	0	X	Data byte will be transmitted.
		No SSDAT action	1	0	0	X	Repeated START will be transmitted.
		No SSDAT action	0	1	0	X	STOP condition will be transmitted and SSSTO flag will be reset.
		No SSDAT action	1	1	0	X	STOP condition followed by a START condition will be transmitted and SSSTO flag will be reset.
38h	Arbitration lost in SLA+W or data bytes	No SSDAT action	0	0	0	X	Two-wire bus will be released and not addressed slave mode will be entered.
		No SSDAT action	1	0	0	X	A START condition will be transmitted when the bus becomes free.

Figure 20-5. Format and State in the Master Receiver Mode

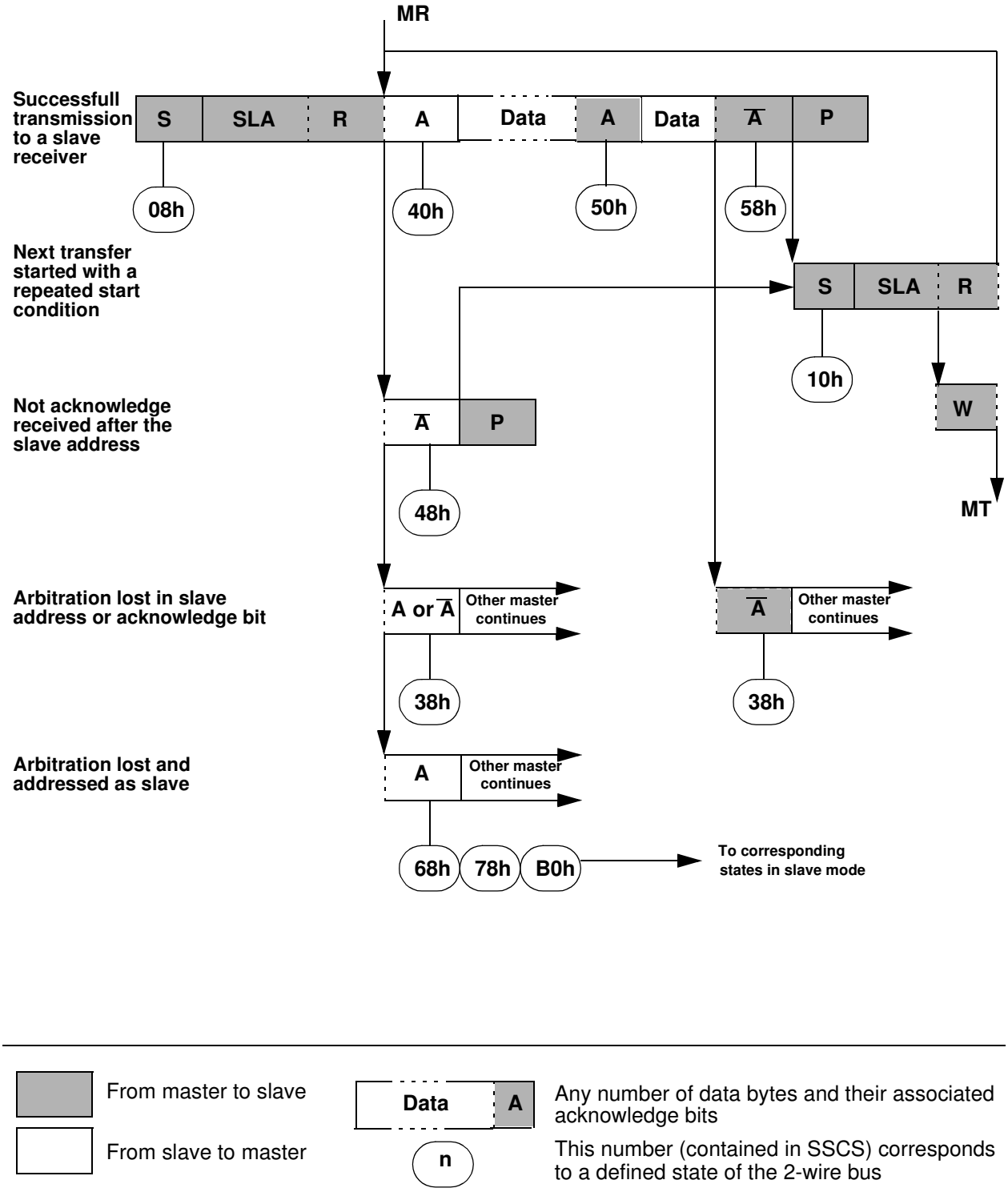


Table 20-6. Status in Master Receiver Mode

Status Code SSSTA	Status of the Two-wire Bus and Two-wire Hardware	Application software response					Next Action Taken by Two-wire Hardware
		To/From SSDAT	To SSCON				
			SSSTA	SSSTO	SSI	SSAA	
08h	A START condition has been transmitted	Write SLA+R	X	0	0	X	SLA+R will be transmitted.
10h	A repeated START condition has been transmitted	Write SLA+R	X	0	0	X	SLA+R will be transmitted.
		Write SLA+W	X	0	0	X	SLA+W will be transmitted. Logic will switch to master transmitter mode.
38h	Arbitration lost in SLA+R or NOT ACK bit	No SSDAT action	0	0	0	X	Two-wire bus will be released and not addressed slave mode will be entered.
		No SSDAT action	1	0	0	X	A START condition will be transmitted when the bus becomes free.
40h	SLA+R has been transmitted; ACK has been received	No SSDAT action	0	0	0	0	Data byte will be received and NOT ACK will be returned.
		No SSDAT action	0	0	0	1	Data byte will be received and ACK will be returned.
48h	SLA+R has been transmitted; NOT ACK has been received	No SSDAT action	1	0	0	X	Repeated START will be transmitted.
		No SSDAT action	0	1	0	X	STOP condition will be transmitted and SSSTO flag will be reset.
		No SSDAT action	1	1	0	X	STOP condition followed by a START condition will be transmitted and SSSTO flag will be reset.
50h	Data byte has been received; ACK has been returned	Read data byte	0	0	0	0	Data byte will be received and NOT ACK will be returned.
		Read data byte	0	0	0	1	Data byte will be received and ACK will be returned.
58h	Data byte has been received; NOT ACK has been returned	Read data byte	1	0	0	X	Repeated START will be transmitted.
		Read data byte	0	1	0	X	STOP condition will be transmitted and SSSTO flag will be reset.
		Read data byte	1	1	0	X	STOP condition followed by a START condition will be transmitted and SSSTO flag will be reset.

Figure 20-6. Format and State in the Slave Receiver Mode

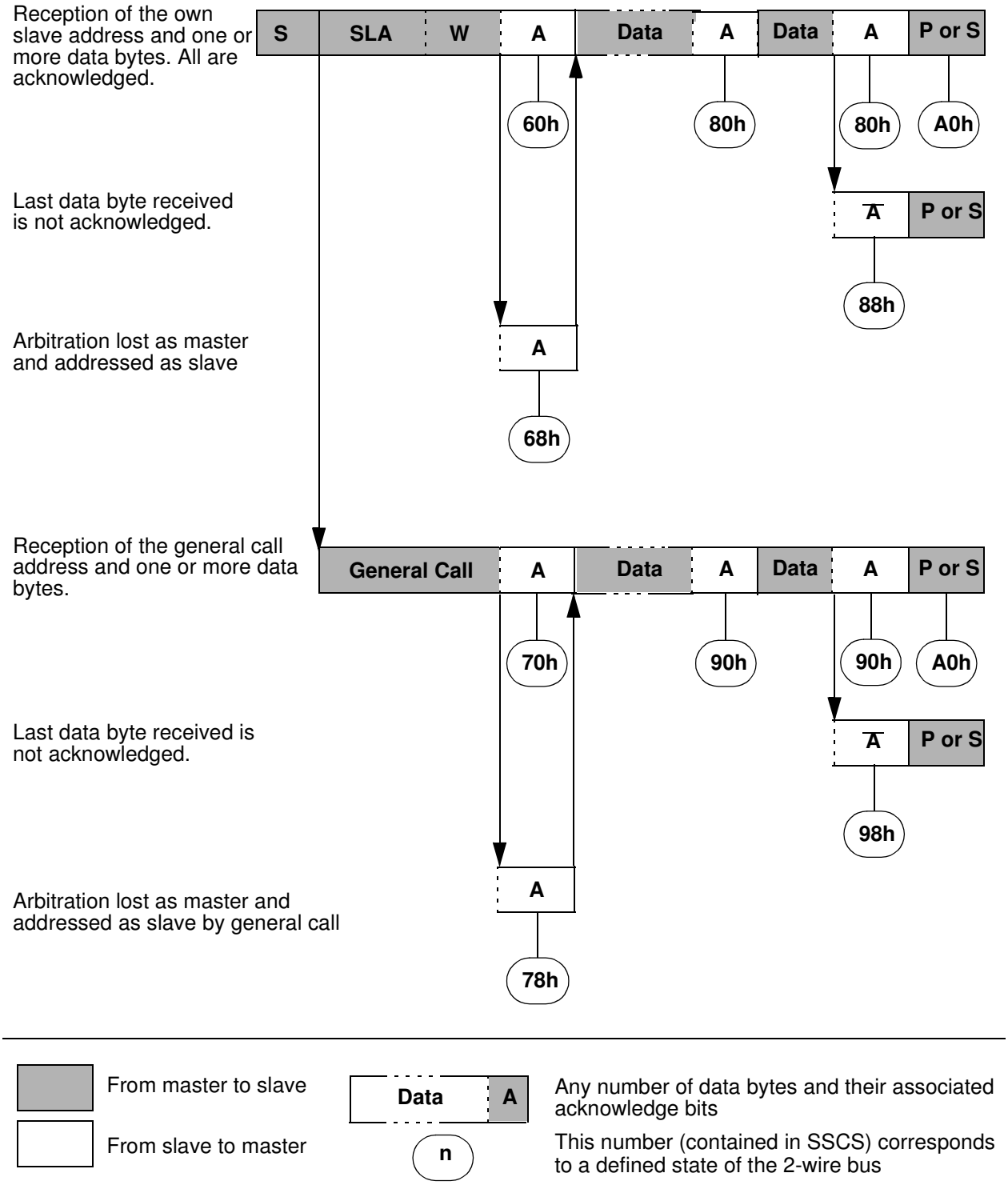


Table 20-7. Status in Slave Receiver Mode

Status Code (SSCS)	Status of the 2-wire bus and 2-wire hardware	Application Software Response					Next Action Taken By 2-wire Software
		To/from SSDAT	To SSCON				
			STA	STO	SI	AA	
60h	Own SLA+W has been received; ACK has been returned	No SSDAT action or	X	0	0	0	Data byte will be received and NOT ACK will be returned
		No SSDAT action	X	0	0	1	Data byte will be received and ACK will be returned
68h	Arbitration lost in SLA+R/W as master; own SLA+W has been received; ACK has been returned	No SSDAT action or	X	0	0	0	Data byte will be received and NOT ACK will be returned
		No SSDAT action	X	0	0	1	Data byte will be received and ACK will be returned
70h	General call address has been received; ACK has been returned	No SSDAT action or	X	0	0	0	Data byte will be received and NOT ACK will be returned
		No SSDAT action	X	0	0	1	Data byte will be received and ACK will be returned
78h	Arbitration lost in SLA+R/W as master; general call address has been received; ACK has been returned	No SSDAT action or	X	0	0	0	Data byte will be received and NOT ACK will be returned
		No SSDAT action	X	0	0	1	Data byte will be received and ACK will be returned
80h	Previously addressed with own SLA+W; data has been received; ACK has been returned	No SSDAT action or	X	0	0	0	Data byte will be received and NOT ACK will be returned
		No SSDAT action	X	0	0	1	Data byte will be received and ACK will be returned
88h	Previously addressed with own SLA+W; data has been received; NOT ACK has been returned	Read data byte or	0	0	0	0	Switched to the not addressed slave mode; no recognition of own SLA or GCA
		Read data byte or	0	0	0	1	Switched to the not addressed slave mode; own SLA will be recognised; GCA will be recognised if GC=logic 1
		Read data byte or	1	0	0	0	Switched to the not addressed slave mode; no recognition of own SLA or GCA. A START condition will be transmitted when the bus becomes free
		Read data byte	1	0	0	1	Switched to the not addressed slave mode; own SLA will be recognised; GCA will be recognised if GC=logic 1. A START condition will be transmitted when the bus becomes free
90h	Previously addressed with general call; data has been received; ACK has been returned	Read data byte or	X	0	0	0	Data byte will be received and NOT ACK will be returned
		Read data byte	X	0	0	1	Data byte will be received and ACK will be returned

Table 20-7. Status in Slave Receiver Mode (Continued)

Status Code (SSCS)	Status of the 2-wire bus and 2-wire hardware	Application Software Response					Next Action Taken By 2-wire Software
		To/from SSDAT	To SSCON				
			STA	STO	SI	AA	
98h	Previously addressed with general call; data has been received; NOT ACK has been returned	Read data byte or	0	0	0	0	Switched to the not addressed slave mode; no recognition of own SLA or GCA
		Read data byte or	0	0	0	1	Switched to the not addressed slave mode; own SLA will be recognised; GCA will be recognised if GC=logic 1
		Read data byte or	1	0	0	0	Switched to the not addressed slave mode; no recognition of own SLA or GCA. A START condition will be transmitted when the bus becomes free
		Read data byte	1	0	0	1	Switched to the not addressed slave mode; own SLA will be recognised; GCA will be recognised if GC=logic 1. A START condition will be transmitted when the bus becomes free
A0h	A STOP condition or repeated START condition has been received while still addressed as slave	No SSDAT action or	0	0	0	0	Switched to the not addressed slave mode; no recognition of own SLA or GCA
		No SSDAT action or	0	0	0	1	Switched to the not addressed slave mode; own SLA will be recognised; GCA will be recognised if GC=logic 1
		No SSDAT action or	1	0	0	0	Switched to the not addressed slave mode; no recognition of own SLA or GCA. A START condition will be transmitted when the bus becomes free
		No SSDAT action	1	0	0	1	Switched to the not addressed slave mode; own SLA will be recognised; GCA will be recognised if GC=logic 1. A START condition will be transmitted when the bus becomes free

Figure 20-7. Format and State in the Slave Transmitter Mode

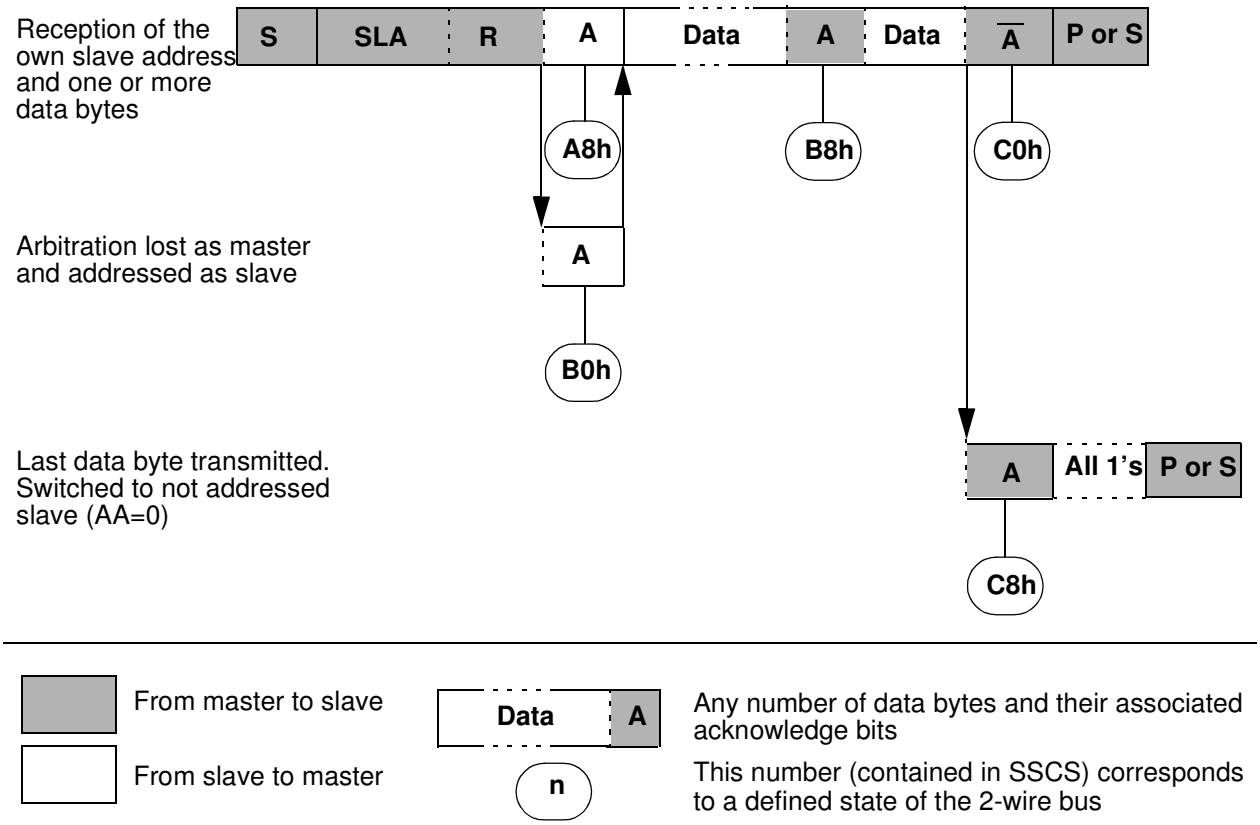


Table 20-8. Status in Slave Transmitter Mode

Status Code (SSCS)	Status of the 2-wire bus and 2-wire hardware	Application Software Response					Next Action Taken By 2-wire Software
		To/from SSDAT	To SSCON				
			STA	STO	SI	AA	
A8h	Own SLA+R has been received; ACK has been returned	Load data byte or	X	0	0	0	Last data byte will be transmitted and NOT ACK will be received
		Load data byte	X	0	0	1	Data byte will be transmitted and ACK will be received
B0h	Arbitration lost in SLA+R/W as master; own SLA+R has been received; ACK has been returned	Load data byte or	X	0	0	0	Last data byte will be transmitted and NOT ACK will be received
		Load data byte	X	0	0	1	Data byte will be transmitted and ACK will be received
B8h	Data byte in SSDAT has been transmitted; NOT ACK has been received	Load data byte or	X	0	0	0	Last data byte will be transmitted and NOT ACK will be received
		Load data byte	X	0	0	1	Data byte will be transmitted and ACK will be received

Table 20-8. Status in Slave Transmitter Mode (Continued)

Status Code (SSCS)	Status of the 2-wire bus and 2-wire hardware	Application Software Response					Next Action Taken By 2-wire Software
		To/from SSDAT	To SSCON				
			STA	STO	SI	AA	
C0h	Data byte in SSDAT has been transmitted; NOT ACK has been received	No SSDAT action or	0	0	0	0	Switched to the not addressed slave mode; no recognition of own SLA or GCA
		No SSDAT action or	0	0	0	1	Switched to the not addressed slave mode; own SLA will be recognised; GCA will be recognised if GC=logic 1
		No SSDAT action or	1	0	0	0	Switched to the not addressed slave mode; no recognition of own SLA or GCA. A START condition will be transmitted when the bus becomes free
		No SSDAT action	1	0	0	1	Switched to the not addressed slave mode; own SLA will be recognised; GCA will be recognised if GC=logic 1. A START condition will be transmitted when the bus becomes free
C8h	Last data byte in SSDAT has been transmitted (AA=0); ACK has been received	No SSDAT action or	0	0	0	0	Switched to the not addressed slave mode; no recognition of own SLA or GCA
		No SSDAT action or	0	0	0	1	Switched to the not addressed slave mode; own SLA will be recognised; GCA will be recognised if GC=logic 1
		No SSDAT action or	1	0	0	0	Switched to the not addressed slave mode; no recognition of own SLA or GCA. A START condition will be transmitted when the bus becomes free
		No SSDAT action	1	0	0	1	Switched to the not addressed slave mode; own SLA will be recognised; GCA will be recognised if GC=logic 1. A START condition will be transmitted when the bus becomes free

Table 20-9. Miscellaneous Status

Status Code (SSCS)	Status of the 2-wire bus and 2-wire hardware	Application Software Response					Next Action Taken By 2-wire Software
		To/from SSDAT	To SSCON				
			STA	STO	SI	AA	
F8h	No relevant state information available; SI= 0	No SSDAT action	No SSCON action				Wait or proceed current transfer
00h	Bus error due to an illegal START or STOP condition	No SSDAT action	0	1	0	X	Only the internal hardware is affected, no STOP condition is sent on the bus. In all cases, the bus is released and STO is reset.

20.3 Registers

Table 20-10. SSCON Register

SSCON - Synchronous Serial Control Register (93h)

7	6	5	4	3	2	1	0
CR2	SSIE	STA	STO	SI	AA	CR1	CR0

Bit Number	Bit Mnemonic	Description
7	CR2	**Control Rate bit 2** See .
6	SSIE	**Synchronous Serial Interface Enable bit** Clear to disable SSLC. Set to enable SSLC.
5	STA	**Start flag** Set to send a START condition on the bus.
4	ST0	**Stop flag** Set to send a STOP condition on the bus.
3	SI	**Synchronous Serial Interrupt flag** Set by hardware when a serial interrupt is requested. Must be cleared by software to acknowledge interrupt.
2	AA	**Assert Acknowledge flag** Clear in master and slave receiver modes, to force a not acknowledge (high level on SDA). Clear to disable SLA or GCA recognition. Set to recognise SLA or GCA (if GC set) for entering slave receiver or transmitter modes. Set in master and slave receiver modes, to force an acknowledge (low level on SDA). This bit has no effect when in master transmitter mode.
1	CR1	**Control Rate bit 1** See Table 20-4
0	CR0	**Control Rate bit 0** See Table 20-4

Table 20-11. SSDAT (095h) - Synchronous Serial Data Register (read/write)

SD7	SD6	SD5	SD4	SD3	SD2	SD1	SD0
7	6	5	4	3	2	1	0

Bit Number	Bit Mnemonic	Description
7	SD7	Address bit 7 or Data bit 7.
6	SD6	Address bit 6 or Data bit 6.
5	SD5	Address bit 5 or Data bit 5.
4	SD4	Address bit 4 or Data bit 4.
3	SD3	Address bit 3 or Data bit 3.
2	SD2	Address bit 2 or Data bit 2.

Bit Number	Bit Mnemonic	Description
1	SD1	Address bit 1 or Data bit 1.
0	SD0	Address bit 0 (R/W) or Data bit 0.

Table 20-12. SSCS (094h) Read - Synchronous Serial Control and Status Register

7	6	5	4	3	2	1	0
SC4	SC3	SC2	SC1	SC0	0	0	0

Bit Number	Bit Mnemonic	Description
0	0	Always zero
1	0	Always zero
2	0	Always zero
3	SC0	**Status Code bit 0** See Table 20-5 to Table 20-9
4	SC1	Status Code bit 1 See Table 20-5 to Table 20-9
5	SC2	**Status Code bit 2** See Table 20-5 to Table 20-9
6	SC3	**Status Code bit 3** See Table 20-5 to Table 20-9
7	SC4	**Status Code bit 4** See Table 20-5 to Table 20-9

Table 20-13. SSADR (096h) - Synchronous Serial Address Register (read/write)

7	6	5	4	3	2	1	0
A7	A6	A5	A4	A3	A2	A1	A0

Bit Number	Bit Mnemonic	Description
7	A7	Slave address bit 7.
6	A6	Slave address bit 6.
5	A5	Slave address bit 5.
4	A4	Slave address bit 4.
3	A3	Slave address bit 3.
2	A2	Slave address bit 2.
1	A1	Slave address bit 1.
0	GC	**General call bit** Clear to disable the general call address recognition. Set to enable the general call address recognition.